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RESEARCH ARTICLE

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Global, regional, and national burden and trends of mental disorders in women of childbearing age: a systematic analysis based on the global burden of disease study 2021

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ABSTRACT

Objectives: This study aims to provide a comprehensive picture of mental disorder burden in women of childbearing age (WCBA), providing scientific evidence for implementing disease prevention and control strategies.

Methods: Based on the data from the Global Burden of Disease study 2021, the age-standardized prevalence rates (ASPR) and disability-adjusted life years (DALY) rates (ASDR) of mental disorders in WCBA, along with their temporal trends, age-period-cohort effects and projection of epidemiological trajectory over the next 10 years were comprehensively analyzed.

Results: In 2021, there were an estimated 343.22 million mental disorder cases in WCBA, with a global ASPR of 17573.61 per 100,000 population and ASDR of 2696.58 per 100,000 population worldwide. Anxiety disorders accounted for the highest prevalence proportion (37.49%), while depressive disorders were the leading DALYs contributors (39.91%) in mental disorders. From 1990 to 2021, the global mental disorder burden showed significant upward trends, with ASPR and ASDR rising at annual average percent changes (AAPC) of 0.35 and 0.38 respectively. The deterioration was more pronounced in anxiety disorders (AAPC 0.74 and 0.73), eating disorders (AAPC 0.42 and 0.41) and depressive disorders (AAPC 0.39 and 0.45). The APC analysis indicated that the disease burden gradually increased with age, and the relative risk peaked during the period of 2017–2021. The projections suggest that the global burden of mental disorders will persistently rise from 2022 to 2031 in WCBA. Specifically, the ASPR and ASDR are projected to reach 23872.35 per 100,000 population and 3570.07 per 100,000 population in 2031, respectively.

Conclusions: The global burden of mental disorders is particularly severe in WCBA and still on the rise. Innovative prevention and healthcare strategies are urgently needed to reduce the burden of highly prevalent, rapidly increasing diseases in high-risk areas.

KEY MESSAGES

1. This study fills a research blank by presenting a comprehensive global, regional and national perspective on the burden, trends and prediction of mental disorders in WCBA from 1990 to 2021.
2. The global burden of mental disorders in WCBA is alarmingly high and continues to rise at an accelerating rate.
3. Urgent implementation of innovative preventive and healthcare strategies is needed to mitigate the burden, especially for disease subtypes with high prevalence or rapid growth, and for regions identified as being at high risk.

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Global burden of disease; women of childbearingage; mental disorders; trend prediction

1. Introduction

Mental disorders are one of the top leading causes of disability worldwide, among which depressive disorders, anxiety disorders, schizophrenia, autism spectrum disorders and other mental disorders ranked the top 25 causes of years lived with disability (YLDs) in 2021 [1]. In 2022, the World Health Organization (WHO) updated the Comprehensive Mental Health Action Plan 2013–2030, which provides guidance and

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suggestions on promoting global mental health. However, in many low and middle income countries, mental health services are plagued by inadequate human and financial resources. According to the estimation of WHO, only 29% of people with mental disorders received proper care, leaving the vast majority completely untreated. Therefore, there is still a long way to go to improve the current situation.

Current evidence points towards substantial sex differences in susceptibility, symptomatology, recurrence and pathophysiology of mental disorders. Women are more vulnerable to certain types of mental disorders [2]. Compared to men, women are twice as likely to be diagnosed with depression and tend to have higher symptom severity, more subjective distress and higher risks of relapse [3,4]. Studies also found that the incidence of anxiety disorder was 1.3–2.4 times higher in women than in men [5]. Gender differences are also evident in eating disorders, with higher proportions observed among adolescent girls and young women [6]. Besides, although the prevalence rates of bipolar disease were comparable between males and females in the general population, females were about 2.4 times more likely to die by suicide than males when in manic episodes [7].

Women of childbearing age (WCBA) have a higher risk of mental disorders, especially during pregnancy and the postpartum period [8–10]. Studies have shown that the most common psychiatric disorders during the perinatal period are depression, anxiety disorders, obsessive compulsive disorder and posttraumatic stress disorder, which promote both pregnancy complications and adverse health outcomes in the offspring [11,12]. A large population-based cohort study in Denmark indicated that women with postpartum psychiatric disorders had a 33% reduction in the rate of a second live birth, compared to those without postpartum psychiatric disorders [13]. Besides, women with poor preconception mental health were 1.48 times more likely to have a non-live birth, and 1.99 times more likely to give birth to a low birthweight baby [14]. Consequently, perinatal mental disorders exert a substantial negative impact on human fertility rates. Accurate analysis and interpretation of epidemiological data on mental disorders in WCBA are helpful to guide evidence-based prevention strategies, resource allocation and policy formulation. To date, global overview of the burden of mental disorders in WCBA is still lacking, suggesting a need for further research and attention in this special population group. Here, the latest data of mental disorders in WCBA were obtained from the Global Burden of Disease Study (GBD) 2021. We delineated the global, regional and national burden of mental disorders, along with their temporal trends in WCBA from 1990 to 2021. In addition, the age-period-cohort (APC) analysis was employed to evaluate the independent effects of age, period and cohort on the burden of mental disorders, and the Bayesian age-period cohort (BAPC) model was used to project both the numbers and age-standardized rates (ASRs) of mental disorders in WCBA from 2022 to 2031.

2. Methods

2.1. Data sources

The GBD 2021, maintained by the Institute for Health Metrics and Evaluation, is a powerful basis for insights on global health trends and challenges. It provides estimations of 459 health outcomes and risk factors across 204 countries and territories from 1990 to 2021 [15]. Detailed information on the data resources, disease and injury hierarchy and statistical methods have been described previously [1]. The estimated numbers and rates of prevalence and disability adjusted life years (DALYs) of mental disorders and 10 most common types of mental diseases, including anxiety disorders, attention-deficit/hyperactivity disorder (ADHD), autism spectrum disorders, bipolar disorder, conduct disorder, depressive disorders, eating disorders, idiopathic developmental intellectual disability, other mental disorders and schizophrenia, were obtained from GBD database (https://vizhub.healthdata.org/gbd-results/). This analysis specifically focused on the mental health status in WCBA, which is defined by WHO as 15–49 years old.

The GBD world population standard was retrieved and employed for the calculation of ASRs, including the age-standardized prevalence rates (ASPR) and the age-standardized DALY rates (ASDR) [16]. Data on global population estimates from 1950 to 2019 and population forecasts from 2017 to 2100 were utilized to evaluate the effects of age, period and birth cohort and predict the numbers and ASRs of mental disorders in WCBA from 2022 to 2031.

2.2. Ethical statement

No additional ethical approval was required for this study, for the usage of publicly accessible data in the GBD study, which was previously approved by the institutional review board of the Washington University School of Medicine. All data provided by GBD were anonymized and de-identified to protect the privacy and confidentiality of the participants. All data were stored and analyzed exclusively on a password-protected computer under the custody of the principal investigator. Access was strictly limited to the essential research personnel involved in this study. Data were not uploaded to any cloud services or shared networks, ensuring the protection of confidentiality.

2.3. Prevalence and DALYs

DALYs are defined as the sum of life years lost due to premature death (YLLs) and YLDs. YLLs are calculated by subtracting the age at death from the longest possible life expectancy for a person at that age. YLDs are measured by taking the prevalence of the condition multiplied by the disability weight for that condition, where disability weights (ranging from 0 for perfect health to 1 for equivalent to death) reflect the severity of different conditions and are developed through surveys of the general public. The proportional contribution of each mental disorder subtype was calculated by dividing its prevalence (or DALYs) by the total prevalence (or total DALYs) of all mental disorders within each region. The numbers and age-specific rates from 1990 to 2021 were also adopted to evaluate the overall burden and temporal trends of mental disorders in WCBA. ASRs were calculated by direct method, using the following formula:

$$\mathrm{ASR} = \frac{\sum_{i=1}^{N}\left(W_i \times a_i\right)}{\sum_{i=1}^{N} W_i} \times 100{,}000$$

where a_i represents the age-specific rate in the *i*th age group, W_i stands for the number of the world's standard population within the same age group, and N is the total number of age groups.

Moreover, to provide more comprehensive and detailed information, the disease burden in subgroups including regions (global, 5 socio-demographic Index (SDI) regions, 21 GBD regions, and 204 countries and territories), age groups (15–19, 20–24, 25–29, 30–34, 35–39, 40–44 and 45–49 years), and types of diseases was also evaluated.

2.4. Joinpoint regression analysis

To assess the temporal changes in the burden of mental disorders in WCBA, Joinpoint regression models were used to determine the optimal number of joinpoints during the whole period. The annual percentage change (APC), average annual percentage change (AAPC), along with their corresponding 95% confidence intervals (CIs) were estimated for the age-standardized trends in both prevalence and DALYs. The benefit of the model is that it can estimate the number of change-points and capture the nonlinear trends in disease burden, providing more accurate information on dynamic changes in rates [17].

2.5. Age-period-cohort (APC) analysis

APC analysis (http://analysistools.nci.nih.gov/apc/), a web tool developed by the US National Cancer Institute, was employed to estimate the effects of age, period, and birth cohort on the burden of mental disorders [18]. Input data for the APC model include either the prevalent case numbers or DALYs, and the corresponding population numbers, with age intervals equal to period intervals. To be specific, the data were arranged into consecutive 5-year periods spanning from 1992–1996 to 2017–2021, and into successive 5-year age groups ranging from 15–19 to 45–49 years.

The key output metrics of the APC analysis include the longitudinal age curve, period relative risks (RR) and cohort RR. In the analysis, the central age group value was defined as the reference value a_0;

the lower of the two central values of calendar period and birth cohort were defined as the reference values p_0 and c_0, respectively.

2.6. Bayesian age-period-cohort (BAPC) analysis

BAPC models were used to forecast the future disease burden of mental disorders from 2022 to 2031 based on historical data. This method has been proven to offer the highest coverage for analyzing predictions of age-stratified rates [19,20]. The BAPC model extends the traditional generalized linear model (GLM) framework within a Bayesian context, enabling dynamic integration of age, period, and cohort effects. These effects are constrained to evolve smoothly over time through a second-order random walk prior, which incorporates inherent regularization to prevent overfitting. The model supports self-validation through its posterior predictive distribution, thereby reducing dependence on external validation datasets. Owing to its flexibility and robustness in capturing complex temporal trends, the BAPC model is particularly well-suited for long-term forecasting of disease burden trends [21]. The mathematical formulation of this model has been described in previous reports [22,23].

2.7. Statistical analysis

The prevalent case numbers, DALYs and ASRs (per 100,000 population) were expressed as the estimates and their 95% uncertainty intervals (UIs). The GBD estimation process itself handles missing data and quantifies uncertainty through advanced modeling techniques, primarily spatio-temporal Gaussian process regression (ST-GPR). The 95% UIs reflect the uncertainty associated with data sparsity and imputation [1]. All statistical analyses and graphs were performed using the R Studio software (version 4.1.1). A two-sided p value less than 0.05 was considered statistically significant.

3. Results

3.1. Global burden of mental disorders in WCBA

From 1990 to 2021, the global number of cases of mental disorders in WCBA rose from 213.09 million (95%UI: 192.35–235.41) to 343.22 million (95%UI: 310.20–379.22). Correspondingly, The ASPR increased from 16055.46 per 100,000 (95%UI: 14206.47–18109.72) to 17573.61 per 100,000 population (95%UI: 15443.91–19954.73) (Table 1). The ASPR of mental disorders in WCBA exhibited distinct temporal trends, with a relatively gradual increase during 1990–2000 (APC = 0.09, 95% CI: 0.02 to 0.16), a decline during 2000–2019 (APC = −0.21, 95% CI: −0.24 to −0.18), and an accelerated increase during 2019–2021 (APC = 7.19, 95% CI: 6.14–8.25), showing an overall AAPC of 0.35 (95%CI: 0.28–0.42) over the entire period (Table S1). Concurrently, the DALYs of mental disorders in WCBA increased from 31.89 million (95% UI: 23.89–41.62) in 1990 to 52.73 million (95% UI: 39.46–69.16) in 2021, while the ASDR rose from 2411.60 (95% UI: 1745.74–3175.57)] to 2696.58 (95% UI: 1939.77–3570.32) per 100,000 population over the same period (Table 1). A similar temporal trend was observed for the ASDR, with an AAPC of 0.38 (95% CI: 0.23–0.52) from 1990 to 2021 (Table S2).

When stratified by disease types, anxiety and depressive disorders were the most prevalent types of mental disorders in 2021, with ASPRs of 7092.27 (95%UI: 5342.86–9173.97) and 6173.49 (95%UI: 4883.29–7781.76) per 100,000 population, respectively (Table 1). Temporal upward trends were found in anxiety disorders (AAPC = 0.74, 95% CI: 0.58 to 0.90), eating disorders (AAPC = 0.42, 95% CI: 0.39 to 0.44), depressive disorders (AAPC = 0.39, 95% CI: 0.27 to 0.50), conduct disorder (AAPC = 0.18, 95% CI: 0.17 to 0.19), autism spectrum disorders (AAPC = 0.07, 95% CI: 0.07 to 0.08), and bipolar disorder (AAPC = 0.04, 95% CI: 0.03 to 0.05). Conversely, notable downward trends were observed in ADHD (AAPC = −0.44, 95% CI: −0.47 to −0.41) and idiopathic developmental intellectual disability (AAPC = −0.28, 95% CI: −0.29 to −0.27) (Table S1 and Figure S1A). Consistent with the prevalence patterns, anxiety and depressive disorders were the leading contributors to mental disorder-related DALYs, with ASDRs of 844.05 (95%UI: 532.79–1232.57) and 1073.50 (95%UI: 686.74–1562.48) per 100,000 population, respectively (Table 1). The

Table 1. The number, age-standardized prevalence and DALYs rates of mental disorders in WCBA from 1990 to 2021.

	Prevalence, in millions (95% UI)		Age-standardized prevalence rate per 100,000 population (95% UI)		DALYs, in millions (95% UI)		Age-standardized DALYs rate per 100,000 population (95% UI)	
	1990	2021	1990	2021	1990	2021	1990	2021
Disease types								
All mental disorders	213.09(192.35–235.41)	343.22(310.20–379.22)	16,055.46(14,206.47–18,109.72)	17,573.61(15,443.91–19,954.73)	31.89(23.89–41.62)	52.73(39.46–69.16)	2411.60(1745.74–3175.57)	2696.58(1939.77–3570.32)
Anxiety disorders	78.10(64.39–97.27)	138.31(112.66–170.65)	5858.75(4448.43–7566.85)	7092.27(5342.86–9173.97)	9.33(6.26–13.08)	16.45(11.02–22.81)	698.30(442.60–1018.65)	844.05(532.79–1232.57)
ADHD	11.49(8.15–15.71)	13.62(9.71–18.45)	814.84(553.94–1153.25)	712.07(479.19–1007.04)	0.14(0.07–0.22)	0.16(0.09–0.26)	9.81(5.08–16.14)	8.56(4.46–14.20)
Autism spectrum disorders	6.81(5.68–8.10)	10.12(8.45–12.03)	508.72(424.19–605.33)	520.05(434.38–618.38)	1.27(0.87–1.79)	1.87(1.28–2.64)	94.30(64.9–132.96)	96.38(66.00–135.65)
Bipolar disorder	8.75(6.98–10.91)	13.05(10.39–16.25)	660.09(498.57–866.65)	667.89(499.44–886.18)	1.89(1.24–2.78)	2.81(1.82–4.10)	142.30(88.88–214.06)	143.77(89.11–216.63)
Conduct disorder	3.14(2.10–4.69)	3.94(2.62–5.94)	205.73(136.33–312.37)	217.28(143.01–331.07)	0.38(0.19–0.62)	0.47(0.24–0.79)	24.60(12.38–41.24)	25.99(13.00–43.45)
Depressive disorders	72.35(63.63–84.07)	121.24(104.69–142.43)	5545.30(4447.49–6858.90)	6173.49(4883.29–7781.76)	12.45(8.43–17.07)	21.04(14.07–29.06)	948.86(617.06–1369.94)	1073.50(686.74–1562.48)
Eating disorders	6.18(4.60–8.31)	9.66(7.14–12.83)	442.01(295.61–633.66)	502.83(332.22–728.25)	1.31(0.79–2.06)	2.04(1.23–3.19)	93.44(53.21–149.85)	106.05(59.75–171.78)
Idiopathic developmental intellectual disability	18.66(10.45–26.55)	24.20(13.87–34.33)	1365.79(762.89–1948.16)	1254.76(720.49–1779.48)	0.73(0.35–1.25)	1.02(0.50–1.69)	53.52(25.54–91.71)	52.78(25.86–87.58)
Other mental disorders	17.28(13.23–22.03)	27.30(20.88–34.61)	1373.78(1021.79–1787.19)	1370.55(1018.59–1786.04)	1.29(0.82–1.93)	2.03(1.31–3.07)	102.35(63.78–155.97)	102.03(63.51–156.04)
Schizophrenia	4.84(3.87–5.91)	7.54(6.01–9.19)	380.10(293.47–479.71)	379.43(289.80–483.21)	3.11(2.27–4.11)	4.84(3.51–6.37)	244.12(168.33–329.91)	243.47(166.79–331.29)
SDI quintiles								
High SDI	40.88(37.12–44.95)	51.85(47.09–57.47)	17,996.05(15,927.23–20,246.37)	21,504.53(18,836.81–24,418.20)	6.59(4.89–8.51)	8.56(6.37–11.02)	2896.06(2106.12–3803.00)	3552.94(2565.79–4691.50)
High-middle SDI	41.62(37.75–45.75)	50.06(45.27–55.47)	15,040.86(13,331.40–16,957.14)	16,379.30(14,298.75–18,704.16)	6.36(4.77–8.36)	7.79(5.82–10.24)	2299.93(1664.54–3028.04)	2537.69(1821.09–3364.84)
Middle SDI	66.95(60.31–73.71)	105.39(95.22–116.19)	15,167.96(13,419.02–17,134.61)	16,956.78(14,905.01–19,236.01)	9.77(7.25–12.78)	15.87(11.72–20.89)	2225.82(1612.52–2935.38)	2546.93(1830.20–3385.01)
Low-middle SDI	45.68(40.67–50.85)	89.87(80.76–99.37)	17,057.20(14,910.65–19,394.23)	17,853.27(15,609.97–20,321.20)	6.51(4.82–8.49)	13.50(10.09–17.63)	2455.21(1772.45–3253.64)	2688.71(1923.44–3562.39)
Low SDI	17.76(15.78–19.91)	45.78(40.72–51.41)	16,254.74(14,115.78–18,585.64)	16,992.80(14,694.04–19,488.55)	2.63(1.97–3.43)	6.95(5.18–9.08)	2430.81(1747.16–3244.48)	2599.72(1865.89–3475.04)

DALYs: disability adjusted life years; WCBA: women of childbearing age; ADHD: attention deficit hyperactivity disorder; SDI: socio-demographic index.

temporal trends of DALYs mirrored those of prevalence over the entire study period (Figure S1B and Table S2).

3.2. Geographic regional burden of mental disorders in WCBA

The burden of mental disorders in WCBA throughout the 21 GBD regions varied in 2021. Depressive and anxiety disorders were the predominant contributors, accounting for 63.22-78.83% of all prevalent cases. Regionally, Central Sub-Saharan Africa carried the highest proportion of depressive disorders (46.50%), while Tropical Latin America showed the highest rates for anxiety disorders (53.06%) (Figure S2A). The highest ASPR was observed in Tropical Latin America [27212.67 (95% UI: 23440.92–31173.58) per 100,000 population]. From 1990 to 2021, significant upward trends in the ASPR of mental disorders were observed in most GBD regions, and the most pronounced increase occurred in Tropical Latin America (AAPC = 0.93, 95% CI: 0.75–1.11). East Asia was the only region that showed a declining trend (AAPC = −0.21, 95% CI: −0.27 to −0.16) (Table S3). Likewise, depressive and anxiety disorders were the predominant contributors, comprising 58.55-78.05% of mental disorder-related DALYs. Regionally, depressive disorders contributed most to mental disorder-related DALYs in Central Sub-Saharan Africa (54.59%), whereas anxiety disorders constituted the largest share in Tropical Latin America (44.23%) (Figure S2B). The trends in the ASDR closely mirrored those of the ASPR throughout the study period. High-income North America (AAPC = 0.97, 95% CI: 0.71–1.23) experienced the most pronounced increase, while East Asia was the only region that showed a declining trend (AAPC = −0.26, 95% CI: −0.28 to −0.25) (Table S4).

3.3. National burden of mental disorders in WCBA

Across the 204 countries and territories, 55 countries had at least one million prevalent cases, with India, China, the United States of America, Brazil and Indonesia being the top five countries and responsible for 45.40% of total mental disorders prevalent in WCBA in 2021. There were 64 countries with ASPRs above 20,000 per 100,000 population, with Greenland, Portugal, Lebanon, Iran (Islamic Republic of) and Brazil ranking as the top five countries (Table S5 and Figure 1(A)). From 1990 to 2021, 190 countries showed

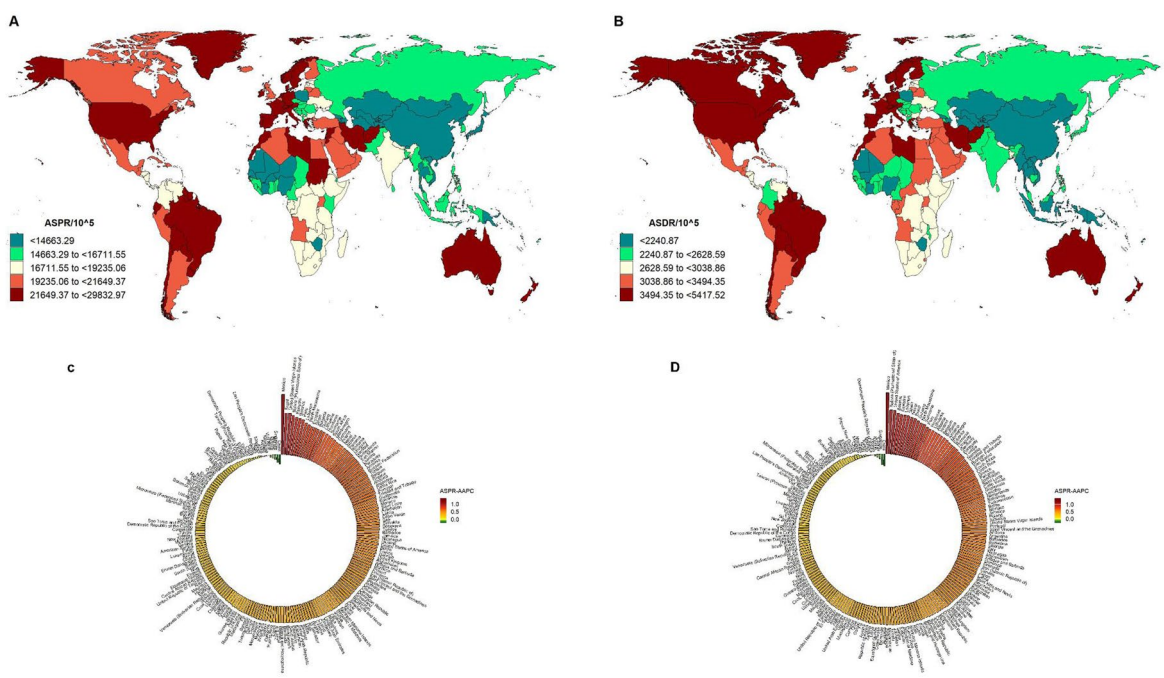

Figure 1. Age standardized prevalence, disability-adjusted life-years (DALYs) rates and average annual percentage change (AAPC) of mental disorders in women of childbearing age (WCBA) in 204 countries and territories between 1990 and 2021. (A) Age standardized prevalence rate (ASPR), (B) age standardized DALYs rate (ASDR), (C) AAPC of ASPR, and (D) AAPC of ASDR.

significant upward trends in prevalence rate, indicating burden homogeneity across most nations, while only Bhutan, Cuba, China, Singapore, and Viet Nam exhibited significant downward trends (Figure 1(C)). The greatest DALYs were found in India, China, United States of America, Brazil, Indonesia and the highest ASDRs were detected in Greenland, Portugal, the United States of America, Lebanon and Greece (Table S6 and Figure 1(B)). Trends in ASDR closely paralleled those in ASPR throughout the study period (Figure 1(D)). Notably, Mexico and China exhibited the highest and lowest AAPC values for both metrics.

3.4. Mental disorders burden in relation to SDI in WCBA

All the SDI regions have experienced a substantial increase both in the prevalence and DALYs of mental disorders in WCBA from 1990 to 2021. In 2021, the middle SDI region had the largest number of cases, reaching 105.39 million (95% UI: 95.22–116.19), followed by the low-middle SDI region [89.87 million (95% UI: 80.76–99.37)]. Whereas, the high SDI region exhibited the highest ASPR [21504.53 (95% UI: 18836.81–24418.20) per 100,000] in 2021, followed by the low-middle SDI region [17853.27 (95% UI: 15609.97–20321.20) per 100,000] (Table 1). All SDI regions showed a similar pattern of changes in prevalence trends of mental disorders, initially smoothing and then accelerating deterioration (Figure 2(A)). Table S7 presents a more detailed analysis in which the data were subgrouped by specific mental diseases and stratified by SDI region. Notably, the rising trends of anxiety and eating disorder were more predominant in the middle SDI region, and the increasing trend of depression was more evident in the high SDI region. In addition, the global decline in ADHD prevalence was confined to low to middle SDI regions, whereas an upward trend persisted in the high SDI region. Likewise, decreasing prevalence trends were also observed for bipolar disorder and idiopathic developmental intellectual disability, though limited exclusively to the low-middle SDI quintile (Table S8). Similar patterns were found in numbers, rates and trends of DALYs. For ASDR, decreasing trends were found for ADHD, schizophrenia, and other mental disorders (all spanning low to middle SDI), autism spectrum disorders (high SDI), bipolar disorder (low-middle SDI), and conduct disorder (middle SDI) (Table S9 and Figure 2(B)).

Figure 2(C) illustrates the long-term trends in the ASPRs and ASDRs for mental disorders across SDI quintile categories by geographical area from 1990 to 2021. Overall, an upward trajectory in the ASPR of mental disorders was observed with increasing SDI values. From 1990 to 2019, ASPRs in most regions exhibited a gradual downward trend with increasing SDI, followed by a sharp rise after 2019. However, the ASPRs in Tropical Latin America and High-income North America exhibited an initially increasing then decreasing trend as SDI values rose before 2019. The correlation between SDI and the ASDR mirrored that with the ASPR.

3.5. Trends by age and disease composition of mental disorder burden in WCBA

Temporal trends in age distribution and disease constitution of prevalence and DALYs of mental disorders in WCBA were also analyzed. The distribution of cases shifted substantially toward older age groups from 1990 to 2021. A consistent decline was observed across all age groups under 30 years, most prominently among the 15 to19-year group, falling from 17.04% to 14.02%. Conversely, all age groups above 30 years showed an increasing proportion, most notably the 45 to 49-year group, which rose from 9.15% to 12.52%. This shift was more pronounced in the middle and high-middle SDI regions (Figure S3A). From 1990 to 2021, the proportion of anxiety disorders among all mental disorders increased by 9.24%, constituting 37.49% of the total prevalence in 2021. In contrast, the share of ADHD decreased by 26.93%, making up 3.69% of all mental disorders in year 2021. Besides, the proportion of idiopathic developmental intellectual disability may relate to economic level, as evidenced by the highest proportion observed in low and low-middle SDI regions. However, these regions have also demonstrated promising downward trends in recent years (Figure S3B). Comparable shifts were observed in the temporal trends of DALYs across these age groups and disease subtypes (Figure S3C,D).

3.6. Age-period-cohort effects on the burden of mental disorders in WCBA

We used an age-period-cohort model to analyze trends in prevalence and DALYs of mental disorders in WCBA across age, time period, and birth cohort. Globally, both of the prevalence and DALY rates of

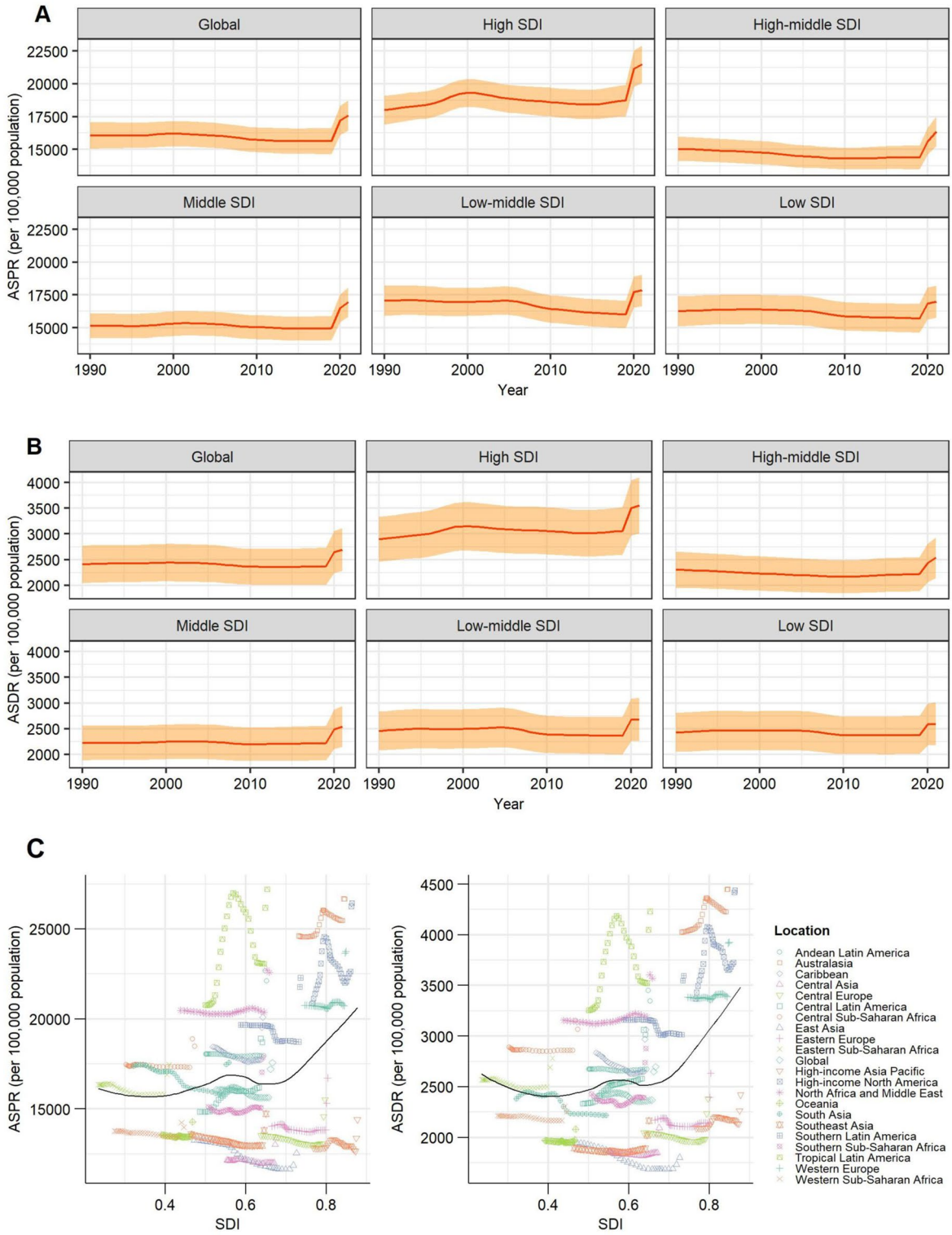

Figure 2. Temporal trends in the burden of mental disorders in WCBA from 1990 to 2021. (A) Trends in ASPR globally and in five SDI regions. (B) Trends in ASDR globally and in five SDI regions. (C) Trends in ASPR and ASDR in 21 GBD regions by SDI. WCBA: women of childbearing age; SDI: socio-demographic Index; ASPR: age-standardized prevalence rate; ASDR: age-standardized DALY rate.

mental disorders increased with age, with the highest found in the 40–44 and 45–49 age groups (Figure 3(A) and S5A). Nevertheless, in the high SDI region, the ASRs initially increased with age and reached its peak at the 25–29 age group, followed by a gradually decreased as age advance. Moreover, the ASRs of ADHD, autism spectrum disorders, other mental disorders and idiopathic developmental

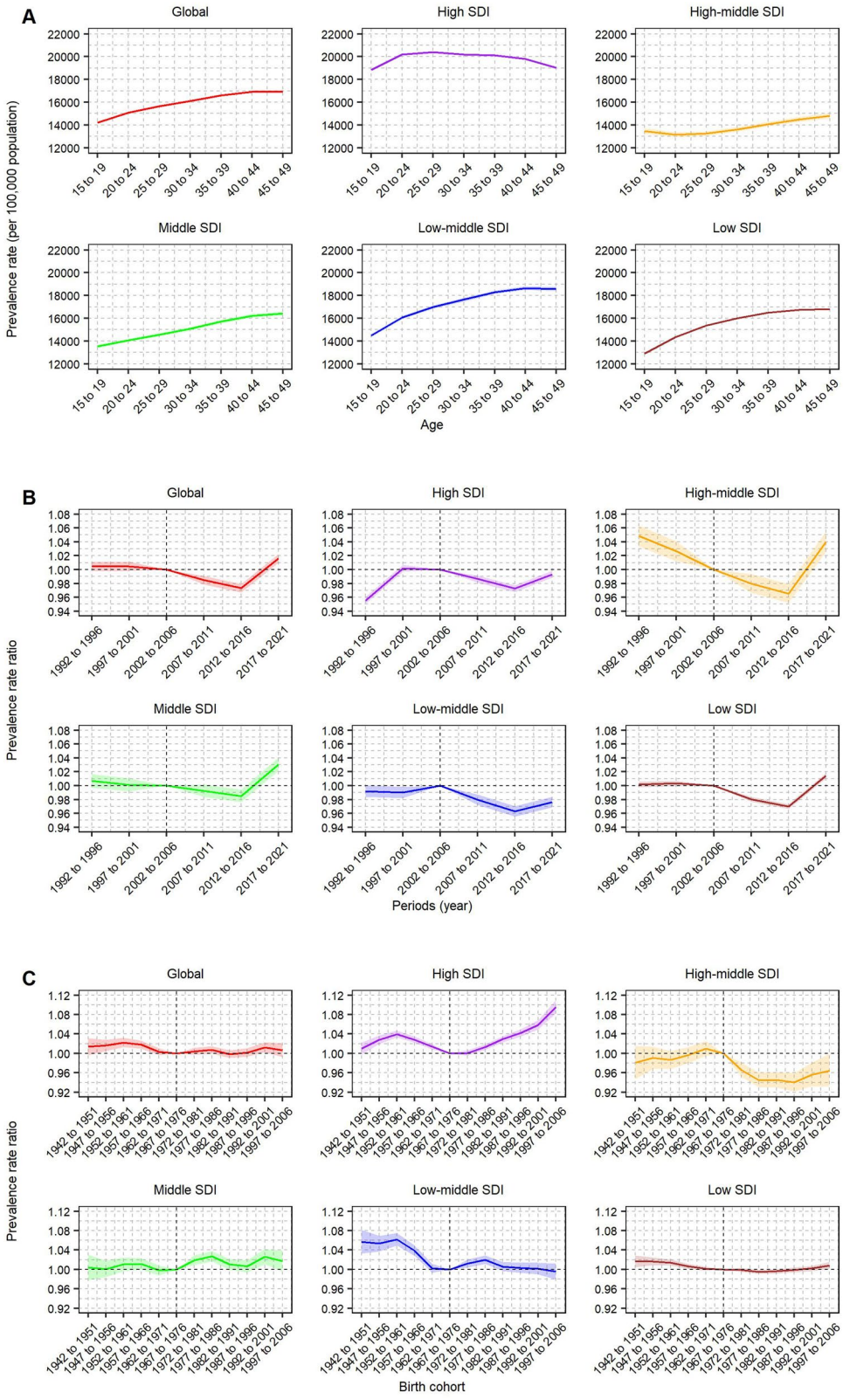

Figure 3. Age, period and birth cohort effects on prevalence of mental disorders in WCBA in global and across five SDI regions from 1992 to 2021. (A) Age effects. (B) Period effects. (C) Birth cohort effects. WCBA: women of childbearing age; SDI: socio-demographic Index; ADHD: attention deficit hyperactivity disorder.

intellectual disability declined with age, whereas the ASRs of anxiety and depressive disorders rose with age. For bipolar disorder, eating disorders and schizophrenia, the peak ASRs were found in the 25–29, 20–24 and 35–39 age groups, respectively (Figures S4A and S6A).

For period effects, the global prevalence and DALYs risk of mental disorders declined from period 1992–1996 to period 2012–2016, but increased afterwards. Similar trends were found in the high-middle, middle and low SDI regions. Notably, all the SDI regions experienced a rapid increase in the risk of prevalence and DALYs during the period of 2017 to 2021 (Figure 3(B) and S5B). Additionally, ADHD and idiopathic developmental intellectual disability demonstrated sustained declines in both prevalence and DALYs risk, whereas eating disorders exhibited persistent upward trajectories throughout the whole period. Concurrently, the risks of anxiety and depressive disorders showed phased elevation during period 2017 to 2021. Significant but relatively stable period effects were identified in other types of mental disorders (Figures S4B and S6B).

For cohort effects, the overall prevalence and DALY risk was relatively stable across the consecutive birth cohorts. Nevertheless, the high SDI region had progressive prevalence and DALYs deterioration in the cohorts born after 1992–2001 (Figure 3(C) and S5C). When stratified by disease types, the risks of ADHD and idiopathic developmental intellectual disability declined in the later birth cohorts, and the risk of eating disorders increased in successively later birth cohorts. Moreover, there were increased risks of anxiety and depressive disorders in cohorts born after 1992–2001. The risks were statistically significant but relatively stable across the birth cohorts in other types of mental disorders (Figures S4C and S6C).

3.7. Predictions of mental disorders burden from 2022 to 2031

Based on the data from 1990 to 2021, the overall burden of mental disorders in WCBA will experience a continuous increase from 2022 to 2031. By 2031, the number of prevalent cases is projected to reach 527.05 million (95% UI: 371.92–682.18), while the ASPR is expected to rise to 23872.35 (95% UI: 17017.91–30726.78) per 100,000 population, representing an increase of 38.32% compared with 2021. When further stratified by types of mental disorders, upward trends are anticipated in anxiety disorders and depressive disorders, downward trends are expected in ADHD, bipolar disorder, eating disorders, other mental disorders and schizophrenia, from 2022 to 2031 (Figure 4 and Table S10). Likewise, the global DALYs due to mental disorders are projected to 80.90 million (95%UI: 50.09–111.71), while the ASDR is expected to rise to 3570.07 (95%UI: 2259.69–4880.45) per 100,000 population, representing a rise of 34.84% relative to 2021. The trends in DALYs were consistent with those in prevalence across different disease types (Figure S7 and Table S10). For other types of mental disorders, stable and smooth trends are expected in the next 10 years.

4. Discussion

This study offers a comprehensive overview of the global burden, trends and projections of mental disorders in WCBA. Our result indicates that more than one-sixth of WCBA in the world live with at least one kind of diagnosable mental disorder, which is substantially higher than that in the general GBD population [1]. There are several factors contributing to the development of mental disorders in WCBA, mainly including genetic factors, physiological factors and social factors. First, it has long been hypothesized that the X-chromosome enriches for genes expressed in the brain, which may exert an impact on intellectual disability and behavior [24]. Second, women are believed to more sensitive to hormones, like estrogen, progesterone, serotonin and melatonin, which fluctuate with menstrual cycle and during pregnancy [25,26]. Fluctuation of these hormone levels can disrupt the function of the brain and finally contribute to mood swings, causing disorders unique to women, such as perinatal depression, premenstrual dysphoric disorder and perimenopausal depression [27]. Third, women are more likely to juggle multiple responsibilities, such as caring for children and the elderly family members, meantime, maintaining full-time jobs and handling the household chores, which cause role overload and can be a major trigger for mental disorders [28].

When stratified by subtypes of mental disorders, anxiety disorders posed the greatest risk to WCBA, followed by depressive disorders, together constituting the largest proportion (75.49%) of mental

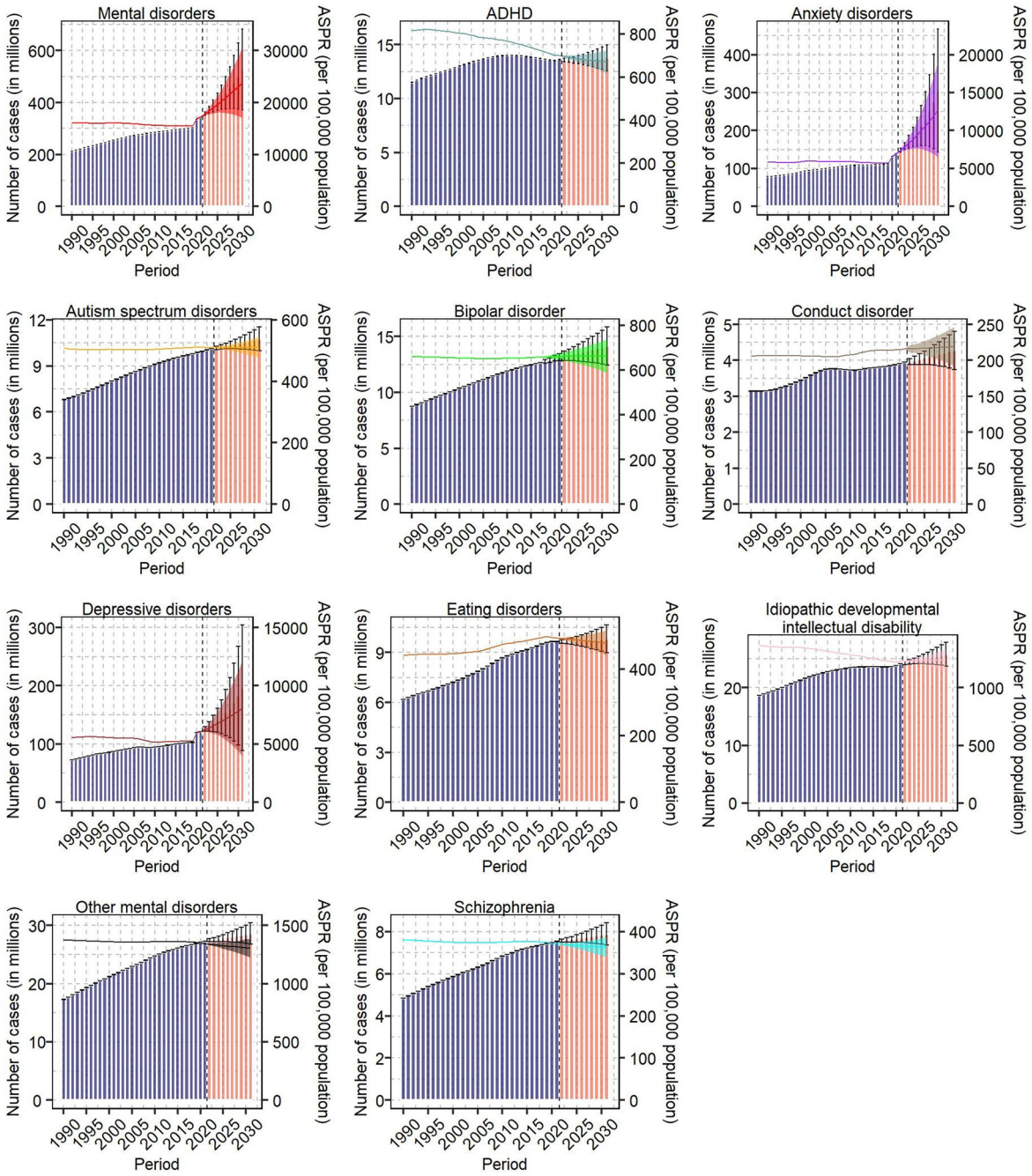

Figure 4. Estimated global case numbers and ASPR trends of mental disorders in WCBA from 1990 to 2031. WCBA: women of childbearing age; ADHD: attention deficit hyperactivity disorder; ASPR: age-standardized prevalence rate.

disorders in WCBA worldwide. Upon stratification by SDI region, we found that the ASPR in WCBA in 2021 was even higher in the high SDI region. There are many possible reasons for such findings. On one hand, the high SDI region possess more risk factors of mental disorders such as increasing lifestyle alternatives, stress, environment changes, social inequality and so on [29,30]. On the other hand, high SDI region tends to possess more social welfare services and medical care resources, which may also promote the inpatient and outpatient consultations of psychological health, and finally facilitate disease diagnosis.

Globally, there was an increasing temporal trends of mental disorders from 1990 to 2021, with a non-negligible steep rise during 2019 and 2021, which was consistent across SDI regions. After stratified by disease types, anxiety and depressive disorders were the primary contributors to this rise. The Corona Virus Disease-19 (COVID-19) pandemic may exert a major impact on mental health both directly and

indirectly [31]. Studies have shown that compared to uninfected controls, levels of brain injury markers, like tTau, GFAP, NfL and UCH-L1, increased in the acute phase of COVID-19 infection, even more prominent in patients with acute phase of neurological dysfunction [32]. A substantial percentage of patients developed long-COVID, accompanied by symptoms such as post-traumatic stress disorder (PTSD), anxiety, or depression 3 and 6 months after the onset of COVID-19 [33]. Also, a range of negative social and economic effects brought by the COVID-19 pandemic might contribute to a rise in these two kinds of mental health problems [34].

Although eating disorder account for a relatively small proportion of mental illnesses, we observed a no negligible prevalence increase during the past few decades in WCBA. When further stratified by SDI region, the most rapid rising trend was found in the middle SDI region (AAPC 1.22), followed by low-middle SDI region (AAPC 0.95), highlighting the severity of the situation in these areas and requiring immediate attention. The reasons for surging of eating disorder in these regions can be attributed to a confluence of sociocultural, economic, and psychological factors. Increased exposure to ideals of thinness and "perfect" body types *via* globalized media platforms has intensified body dissatisfaction, particularly among adolescents and young adults. The swift pace of urbanization has fostered a reliance on processed and high-calorie foods, erratic eating patterns, coupled with sedentary lifestyles, thereby disrupting innate hunger cues and cultivating maladaptive eating behaviors [35]. The stress associated with socio-economic transitions, including job insecurity, financial strain, and social displacement, can also contribute to the development of disordered eating behaviors as coping mechanisms.

Despite a promising temporal downward trend in burden of ADHD worldwide (AAPC −0.44%), the high SDI region showed an upward trend (AAPC 0.24%) differed from other SDI regions. Furthermore, during 2019 and 2021, a notable increase in the burden of ADHD was observed both in the high and high-middle SDI regions. It is widely believed that the prevalence of ADHD has been on the rise following the update of its diagnostic criteria and classification. Researches have shown that the actual prevalence of ADHD in women during pregnancy and breastfeeding is gradually rising in some developed countries [36–38]. Besides, studies carried out in developed countries such as Finland and the US also reported a rise in ADHD diagnoses and overall prevalence during the COVID-19 pandemic, which aligns with the results of our study [39,40]. Interestingly, despite a promising global temporal downward trend in idiopathic developmental intellectual disability (AAPC −0.28%) observed from 1990 to 2021, there were notable upward trends in the high, high-middle and middle SDI regions during the period 2019 to 2021. These results may have multiple underlying causes. The shift of early identification of childhood mental disorder onset from specialized to school and primary care in many countries, along with COVID-19-induced life changes, may have caused inattentive symptoms, especially in girls and young women, and brought these cases to light.

The identification of subgroups with declining trends, though limited, offers crucial insights for global mental health policy. The transient declines observed in certain geographic areas prior to 2021, and the more sustained reductions in disorder-specific burden within particular SDI regions, suggest that the rising global tide of mental disorders is not inevitable. These exceptions can likely be attributed to the positive impact of targeted interventions, health system strengthening, or favorable shifts in social determinants that have, in some contexts, outpaced underlying risk factors. For instance, the declines in ADHD and schizophrenia burden across low to middle SDI regions may partly benefit from the implementation of strategies to reduce the risk exposure worldwide, such as improved perinatal care, recommendation for healthy lifestyles, guidance of nutrition and medication use during pregnancy, policies or laws have been introduced to reduce lead exposures [41–43]. Conversely, in high SDI regions, the observed reduction in ASD burden is primarily attributable to advancements in early diagnosis, evidence-based interventions, and comprehensive support systems [44]. Future research should focus on in-depth case studies of these regions. Identifying the underlying factors that contribute to these favorable outcomes, such as investment in primary care integration, community-based programs, and public health campaigns, could provide a replicable blueprint for other nations aiming to curb the growing burden in WCBA.

In the APC analysis, age effects showed a greater impact on prevalence of mental disorders. Although the global prevalence rate of mental disorders in WCBA gradually increased with age, the high SDI region showed a different pattern. The prevalence rate in this region reached its peak in the 25–29-year age

group, followed by a gradual decline. There may some potential factors contributed to the younger onset of mental illnesses in this region. First, in economically developed countries, young people often face significant study and work pressure, which can stem from various sources, such as the need to succeed in their careers and achieve academic excellence. Second, evidence have suggested that social media and digital technology overload were associated with mental health among young people, and the usage rate of social media is particularly high in the developed countries [45]. Third, emerging evidence suggests that weakening social cohesion driven by urbanization, the prevalence of nuclear families, and the reduction in face-to-face interactions in developed countries also constitutes a risk factor for mental disorders among youth. In addition, the worsen prevalence of anxiety and depressive disorders was mainly driven by period effects, while the rising prevalence of eating disorders and the falling prevalence of ADHD and idiopathic developmental intellectual disability were attributed to a combination of period and cohort effects. These findings support our earlier hypotheses.

The BAPC model predicted that the burden of mental disorders in WCBA would experience a rapid increase over the next 10 years worldwide, especially in anxiety and depressive disorders. The deteriorating mental health condition is due to complicated situations, among which economic and social circumstances may be essential. As mentioned in the latest 2025 World Economic Situation and Prospects (WESP) report that the global economy is at a critical juncture, grappling with interconnected challenges that include the scarring effects of the COVID-19 pandemic, ongoing conflicts, high levels of public debt, economic and social inequalities, and the climate crisis. Consequently, the unfavorable economic situation will contribute to the rise in unemployment, which can easily trigger mental problems such as anxiety and depression. The high prevalence rate of mental illness poses challenges not only to mental health system but also to social stability. Considering the serious situation, effective preventive and therapeutic strategies are urgently needed. First, build a tiered mental health system to provide continuous and equitable care. This involves strengthening psychiatric hospitals for complex cases, expanding community centers for rehabilitation, and integrating mental services into primary care for early intervention. Telepsychiatry and mobile clinics should be enhanced to increase accessibility in underserved areas. Second, enhance medical insurance coverage for mental healthcare by including long-term therapy, essential medications, and follow-up consultations. Increase reimbursement rates for both outpatient and inpatient services, streamline insurance claims processes, and provide targeted subsidies for low-income patients with severe conditions. These steps will alleviate the financial burden on individuals and ensure continuous access to effective treatment. Third, it is essential to improve the accessibility of psychological guidance and rehabilitation services by providing psychoeducation and targeted support to family members to facilitate early family-based interventions. At the same time, the expansion of public mental health services, which include community-based social interventions, online psychological counseling, and psychological support hotlines, must be prioritized [46,47]. Fourth, strengthen legal safeguards by enacting anti-discrimination laws and paid mental health leave. Integrate gender-sensitive protocols into healthcare systems and expand coverage for perinatal support, trauma-informed therapy, and culturally adapted counseling. Establish community programs to provide crisis intervention and psychosocial aid for vulnerable women [48,49]. Finally, fostering communication, cooperation, and coordination among countries and international organizations is essential to enhance capacities for responding to mental health crises. High-income countries and international agencies such as the World Health Organization (WHO) and the United Nations Population Fund (UNFPA) should offer financial support and technical assistance to low- and middle-income countries to facilitate the development, implementation, and monitoring of national mental health strategies. This should include sharing best practices, providing training materials, and supporting context-specific intervention research.

There are some strengths of this study. First, to the best of our knowledge, this is the first study providing a comprehensive estimation and prediction on the burden of mental disorders in WCBA, highlighting an urgent need for further attention to this population group. Second, the in-depth analysis stratified by disease types and regions offers a deeper understanding of disease temporal trends, providing valuable insights for policy-making that target the highly prevalent and rapidly increasing diseases, as well as the high-risk populations and areas. Third, the Joinpoint regression was employed to analyze the temporal trend of mental disorders, which takes into account the trend transitions during the whole period. This approach

allowed us to identify multiple Joinpoints and gain deeper insights into the factors driving their prevalence. There are some limitations of this study. First, due to the stigma attached with mental disorders and varied health care levels across regions and countries, there may be potential underestimation of disease prevalence. Specifically, the less developed countries may report lower disease burdens due to poorer medical condition and limited access to health care. Second, while the GBD's use of diverse data sources (including censuses, disease registries, and scientific literature) is a major strength, the heterogeneity in methods, quality, and coverage across these sources introduces additional uncertainty into the estimates. The 95% UIs account for this, meaning our findings should be interpreted with greater caution for geographies that rely heavily on modeled data rather than direct empirical evidence. Third, our predictions of future burden are statistical projections based on past data and demographic trends. They do not explicitly model the complex influence of future socioeconomic, environmental, or ecological factors. As such, these estimates could be influenced by unforeseen events, policy changes, or innovations not captured in the model.

5. Conclusions

In summary, an unfavorable overall temporal trend of mental disorders burden in WCBA was detected, the increasing trends were observed across all SDI regions and most of the GBD regions and countries, putting new challenges to the healthcare system and policy-making with regard to mental health in WCBA. Notably, the global healthcare of mental disorders in WCBA needs to be tilted towards the anxiety and depressive disorders, which account for the largest proportion of mental disorder and are still on the rise. Besides, more attention should be paid to eating disorder in the middle and low-middle SDI regions due to its alarming increasing trend. To address these challenges and advance the objectives, we recommend implementing a coordinated strategy that includes establishing integrated and tiered mental health services, improving financial protection through insurance reform, expanding community-based and digital interventions, strengthening legal safeguards against discrimination, and enhancing international cooperation for resource sharing and capacity-building. Sustained and context-sensitive investment in mental health infrastructure and services is essential to effectively reduce the burden of mental disorders in this vulnerable population.

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Ethical approval

This study was reviewed by the institutional review board of Shanghai Mental Health Center, and an ethical exemption was granted (exemption number: 2025 M-01). This study was conducted in accordance with Declaration of Helsinki (as revised in 2013) and Article 32 of China's Ethical Review Measures for Biomedical Research Involving Humans (2023).

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data were obtained through an online query tool from the website of GBD 2021 (https://vizhub.healthdata.org/gbd-results/), and no permissions were required to access the data.

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